Pricing Models & Modeling TCO

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Outline

Pricing Models

Cost Metrics

Total Cost of Ownership (TCO)

Pricing

Providers

How do providers price cloud resources?

Consumers

What is the cost of using cloud resources?

Pricing Models

Price

Unit cost of using cloud resources, e.g., based on time, number of transactions, etc.

Charge by per hour, per minute, per second

Factors

Overhead in design, development, deployment, and operation of cloud services, and other IT resources

Opportunities to reduce expenses via IT resource sharing and data center optimisation

Market competition and regulatory requirements

Cost metrics and associated prices – depends on **on-demand** (pay-per-use) or **reserved allocation** (upfront fee + discounted hourly rate)

Discount for higher usage/volumes

Payment options and schedules

- Customisation
- Prepaid, postpaid, instalments

Examples

IaaS

Pricing based on resource allocation and usage including number of virtual servers, amount of network data transferred, and allocated storage capacity

PaaS

Price also depends on software configurations, development tools and any licensing fees

SaaS

Price is determined by number of application modules in the subscription, number of nominated cloud service consumers, and number of transactions

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AWS EC2 (https://aws.amazon.com/ec2/instance-types/)

Virtual machines (IaaS) are divided into many types to support different types of applications and consumer requirements

Types

- General purpose – M9, M8, M7, T8, ...
- Compute optimized – C9, C8, C7, C6, ...
- Memory optimized – R9, R8, X8, X2, ...
- Accelerated computing – P6, P5, G7, G6, ...
- Storage optimized – I8, I7, H1, ...
- HPC optimized – Hpc8, Hpc7, Hpc6, ...

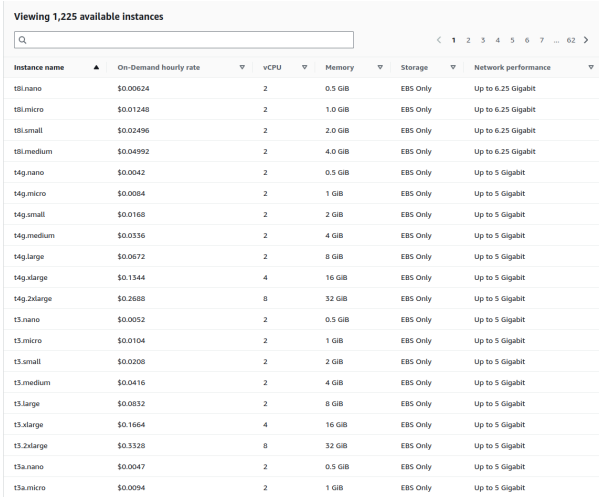
AWS pre-created many types of machines

Each instance type is further divided into subtypes based on varying combinations of CPU, memory, storage, and network capacity

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AWS EC2

Pricing



Viewing 1,225 available instances

1 2 3 4 5 6 7 ... 62

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t8i.nano	$0.00624	2	0.5 GiB	EBS Only	Up to 6.25 Gigabit
t8i.micro	$0.01248	2	1.0 GiB	EBS Only	Up to 6.25 Gigabit
t8i.small	$0.02496	2	2.0 GiB	EBS Only	Up to 6.25 Gigabit
t8i.medium	$0.04992	2	4.0 GiB	EBS Only	Up to 6.25 Gigabit
t4g.nano	$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t4g.micro	$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t4g.small	$0.0168	2	2 GiB	EBS Only	Up to 5 Gigabit
t4g.medium	$0.0336	2	4 GiB	EBS Only	Up to 5 Gigabit
t4g.large	$0.0672	2	8 GiB	EBS Only	Up to 5 Gigabit
t4g.xlarge	$0.1344	4	16 GiB	EBS Only	Up to 5 Gigabit
t4g.2xlarge	$0.2688	8	32 GiB	EBS Only	Up to 5 Gigabit
t3.nano	$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.micro	$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.small	$0.0208	2	2 GiB	EBS Only	Up to 5 Gigabit
t3.medium	$0.0416	2	4 GiB	EBS Only	Up to 5 Gigabit
t3.large	$0.0832	2	8 GiB	EBS Only	Up to 5 Gigabit
t3.xlarge	$0.1664	4	16 GiB	EBS Only	Up to 5 Gigabit
t3.2xlarge	$0.3328	8	32 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	$0.0047	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	$0.0094	2	1 GiB	EBS Only	Up to 5 Gigabit

https://aws.amazon.com/ec2/pricing/on-demand/

Example

AWS EC2 Pricing

Charges differ across region and depends on the operating costs (energy, communication, and maintenance)

for spiky workloads

On-demand – based on usage in terms of time; charge to nearest second with minimum of 60 seconds

for steady workloads

Reserved – onetime upfront cost + pay as your go (cost is lower than on-demand)

Spot – consumers bid for unused capacity at a lower price than on-demand rate; suitable for running jobs that are not time critical and at a lower cost

for non-critical workloads

Savings Plan – flexible pricing model that offers low prices in exchange for a commitment to a consistent usage for a longer term

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you decide based on your workload!

SaaS Pricing Models

Integrated Pricing

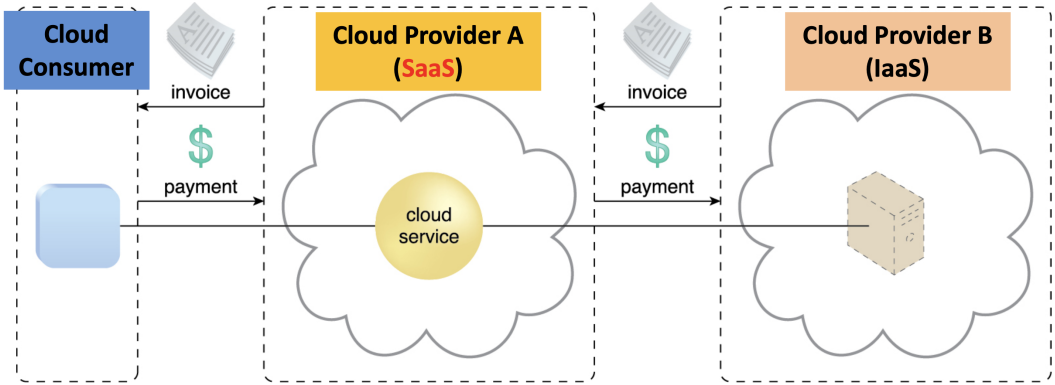
Consumer pays SaaS provider

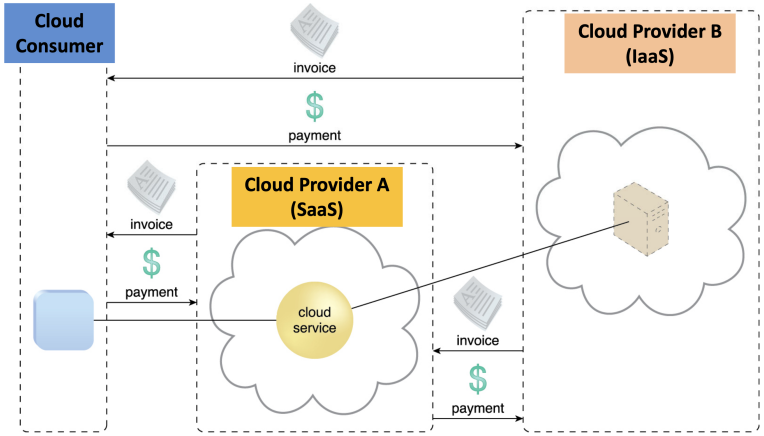
SaaS provider in turn pays IaaS provider for hosting the SaaS service

Separate Pricing

Consumer pays SaaS provider for SaaS service and IaaS provider for hosting the SaaS service

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Cloud Consumer
invoice
$
payment
Cloud Provider A (SaaS)
cloud service
invoice
$
payment
Cloud Provider B (IaaS)


Cloud Consumer
invoice
Cloud Provider B (IaaS)
payment
Cloud Provider A (SaaS)
invoice
payment
cloud service
invoice
payment

Cost Metrics

Financial analysis of cloud adoption

Compare on-premise (total cost of ownership (TCO)) vs cloud-based provisioning

Usage Scenario

Cloud consumer

Cloud provider – how to price a cloud service, etc.

Business Cost Metrics

Business costs = capital expenses + operational expenses

Up-front costs

- capital expenses (Capex)
- initial investment to fund IT resources
- obtain resource, deploy and administer
- on-premise is high – e.g., hardware, software and deployment costs
- cloud based is low – e.g., labour cost for setting up cloud environment

On-going costs

- operational expenses (Opex)
- run and maintain IT resources – e.g., licensing fee, electricity, maintenance
- higher for on-premise

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cost of borrowing

Cost of capital

Cost incurred to raise funds – if high, leasing cloud-based is justified

Sunk costs

Prior investment on existing IT resources

If significant, cannot justify cloud-based

Integration costs

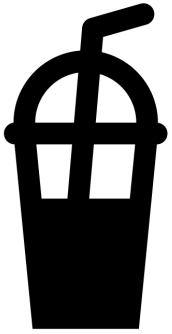
Effort needed to inter-operate IT resources on new environment – e.g., new cloud platform

High cost $\Rightarrow$ cloud-based less appealing

Locked-in costs

Movement from one cloud provider to another – high cost $\Rightarrow$ cloud-based less appealing

Take a Break







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Cloud Usage Cost Metrics

Network usage (data transfer)

Inbound, outbound and intra-cloud network traffic

Server usage (compute)

On-demand and reserved virtual machine allocation

Cloud storage device (storage)

On-demand storage allocation, I/O data transfers

Cloud service (service)

Subscription duration, number of users, number of transactions

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Network Usage Metrics

Total network traffic in bytes

Accumulated over a pre-defined period – e.g., up to 1GB free, S$ 0.01/GB till 1TB and S$ 0.005/GB after 1TB per month

Inter-Cloud

- Data replication, synchronization
- May have different rates for in-bound and out-bound (many providers do not charge for in-bound)

Intra-Cloud

Some providers may not charge for it

because that's necessary for the cloud to work

Server Usage Metrics

Range of performance attributes – CPU, RAM, dedicated storage, ...

On-demand

Pay-per-usage fee short-term

Cumulative from server start to stop date

Reserved

Upfront cost for reservation

Start date to expiry date

Costs: daily/monthly/yearly

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Cloud Storage Device Metrics

On-demand storage space

Size of storage space in bytes

Cumulative sum over date of allocation to release

I/O data transferred metric

I/O data in bytes

Some providers may not charge

Cloud Service Usage Metrics

Application subscription duration

Cumulative sum from subscription start to expiry date

Number of nominated users

Number of users – e.g., S$ 0.90/additional user per month

Number of transactions

Number of request-response message exchanges

E.g., S$ 0.05 per 1,000 transactions

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Total Cost of Ownership of a Datacenter I

Costs

capital expenses (Capex) + operational expenses (Opex)

Capex

Upfront investment depreciated over a certain time frame such as construction cost of datacenter and purchase price of servers

Opex

Recurring cost of running datacenter such as electricity costs, repairs and maintenance, salaries, etc.

Total Cost of Ownership of a Datacenter II

TCO

Cost of owning the hardware resources (hardware + infrastructure to keep hardware working)

`datacenter depreciation` + building
`datacenter Opex` +
`server depreciation` + IT equipment
`server Opex`

Comparing TCOs

Case A

High-end servers

Case B

Lower-cost, higher-power servers

on average case B lowest cost
because cost of higher power doesn't
offset cost of depreciation

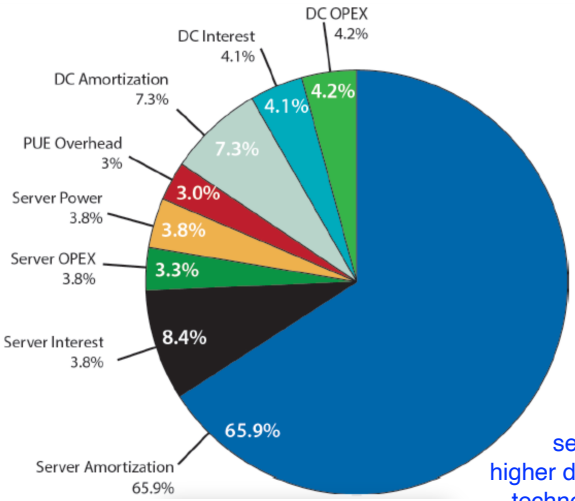
Case C

Case B with 50% datacenter utilisation

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Case A

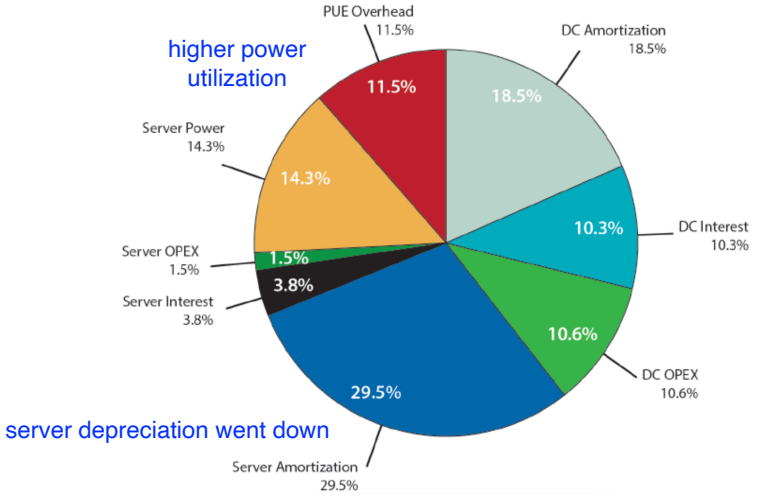
Cost Distribution



server amortization:
higher depreciation due to rapid technological obsolescence

Case B

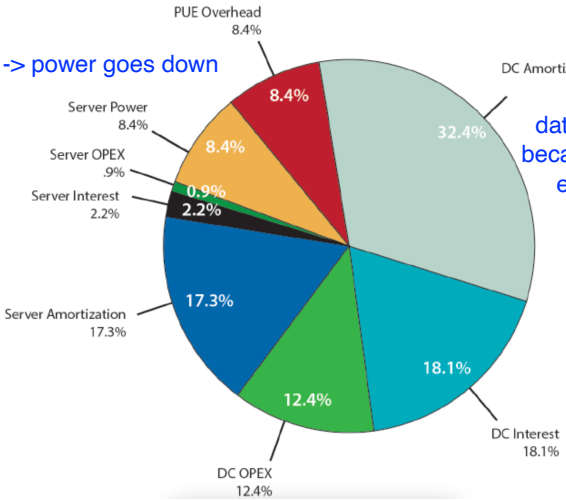
Cost Distribution



Case C

Cost Distribution

50% utilization -> power goes down



data center depreciation goes up because you're only using 50% of it even though it has been built

Test

Date – **November 3rd, 2026**

Time – **18:30 – 20:00**

Venue – LT 17

Format – Examplify (Make sure your laptops are **charged** and can last the time!)

Open-book

- We will be using the **Non-secure Block Internet** option on Examplify

Consists of MCQs, MRQs and structured questions

- Few book work
- More about applying your knowledge ☺

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Next Week – Guest Lecture

EVENT DETAILS

27 October 2026

6:30pm – 8:00 pm
NUS, LT 17



Eugene Swee

Principal AI Engineering Solutions Architect
APJ AI Innovation Hub,
Dell Technologies

"Over 20 years of experience across AI Engineering, Software Development, Operations, DevOps and MLOps, Presales and post-sales."

GUEST LECTURE

From Cloud Native to AI-Native: Building Modern Applications

What you'll explore

In today's fast-paced digital landscape, technology is a key differentiator, helping businesses deliver value faster, smarter, and at scale. This session explores how Cloud Native approaches create agile, resilient applications and infrastructure ready for AI-driven innovation.

Join Dell Technologies for an insightful session on the transformative shift in software and AI Engineering. Learn about how **microservices**, **AI Infrastructure**, **MLOps**, and **AI Engineering** work together to build, deploy, and manage modern intelligent applications.

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DELL Technologies

References

Chapter 15, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, et al., Prentice-Hall, 2013

Chapter 9, Essentials of Cloud Computing, CRC Press, 2015

Chapter 6, Modeling Costs in "The Datacenter as a Computer", 2019

Continuous Feedback



https://forms.cloud.microsoft/r/ZJxTf2wFjr

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